Spatial Discretization Methods for the Neutron Transport Equation in 1D Slab Geometry

Dmitriy Y. Anistratov

Department of Nuclear Engineering
North Carolina State University

NE 723
Neutron Transport Theory
Fall 2026

Outline

1. One-group transport problems in 1D slab geometry
2. Method of discrete ordinates (S_N)
3. Method of characteristics
   - Concept of particle balance in discrete space
   - Step characteristics
   - Linear characteristics
4. Diamond differencing scheme
5. Linear discontinuous Galerkin method
6. Upstream corner-balance method
7. Source iterations for one-group transport problems

Transport problems for 1D slab geometry

We consider steady-state one-group transport equation for problems in 1D slab geometry with isotropic scattering:

$$\mu \frac{\partial \psi(x,\mu)}{\partial x} + \Sigma_t(x)\psi(x,\mu) = \frac{1}{2}\Big(\Sigma_s(x) + \nu_f(x)\Sigma_f(x)\Big)\int_{-1}^{1} \psi(x,\mu')d\mu' + \frac{1}{2}q(x)\,, \quad (1)$$

$$x \in [0,X]\,, \quad \mu \in [-1,1]$$

$$\psi(0,\mu) = \psi_{in}^{+}(\mu) \quad \text{for } \mu > 0\,, \quad \psi(X,\mu) = \psi_{in}^{-}(\mu) \quad \text{for } \mu < 0\,.$$

The scalar flux and current are defined by

$$\phi(x) = \int_{-1}^{1} \psi(x,\mu)d\mu\,, \quad J(x) = \int_{-1}^{1} \mu\psi(x,\mu)d\mu\,.$$

Transport problems for 1D slab geometry

The k-eigenvalue problem for one-group transport equation for 1D slab geometry:

$$\mu \frac{\partial \psi(x,\mu)}{\partial x} + \Sigma_t(x)\psi(x,\mu) = \frac{1}{2}\left(\Sigma_s(x) + \frac{1}{k}\nu_f(x)\Sigma_f(x)\right)\int_{-1}^{1}\psi(x,\mu')d\mu' \,, \qquad (2)$$

$$x \in [0, X] \,, \quad \mu \in [-1, 1]$$

Vacuum boundary conditions

$$\psi(0,\mu) = 0 \quad \text{for } \mu > 0 \,, \quad \psi(X,\mu) = 0 \quad \text{for } \mu < 0 \,.$$

Reflective boundary conditions

$$\psi(0,\mu) = \psi(0,-\mu) \quad \text{for } \mu > 0 \,, \quad \psi(X,\mu) = \psi(X,-\mu) \quad \text{for } \mu < 0 \,.$$

This is the eigenvalue problem for the multiplication factor k.

Transport problems for 1D slab geometry

The one-group transport problem for critical size in 1D slab geometry:

$$\mu \frac{\partial \psi(x,\mu)}{\partial x} + \Sigma_t(x)\psi(x,\mu) = \frac{1}{2}\Big(\Sigma_s(x) + \nu_f(x)\Sigma_f(x)\Big) \int_{-1}^{1} \psi(x,\mu')d\mu' \,, \qquad (3)$$

$$x \in [0, X] \,, \quad \mu \in [-1, 1]$$

Vacuum boundary conditions

$$\psi(0,\mu) = 0 \quad \text{for } \mu > 0 \,, \quad \psi(X_{crit},\mu) = 0 \quad \text{for } \mu < 0 \,.$$

Reflective boundary conditions

$$\psi(0,\mu) = \psi(0,-\mu) \quad \text{for } \mu > 0 \,, \quad \psi(X_{crit},\mu) = \psi(X,-\mu) \quad \text{for } \mu < 0 \,.$$

This is the eigenvalue problem for critical size X_{crit} of the physical system.

Method of discrete ordinates (S_N)

We introduce the quadrature set

$$\{\mu_m, w_m, m = 1, ..., M\}$$

$$\sum_{m=1}^{M} w_m = 2\,, \quad \sum_{m=1}^{M} \mu_m w_m = 0\,.$$

The *discrete-ordinates* method applied to the transport equation (3) is given by

$$\mu_m \frac{d\psi_m(x)}{dx} + \Sigma_t(x)\psi_m(x) = \frac{1}{2}\Big(\Sigma_s(x) + \nu_f(x)\Sigma_f(x)\Big) \sum_{m'=1}^{M} \psi_{m'}(x) w_{m'} + \frac{1}{2} q(x)\,, \quad (4)$$

$$\psi_m(0) = \psi_{in}^{+}(\mu_m) \quad \text{for } \mu_m > 0\,, \quad \psi_m(X) = \psi_{in}^{-}(\mu_m) \quad \text{for } \mu_m < 0\,,$$

where

$$\psi_m(x) = \psi(x, \mu_m)\,,$$

$$\phi(x) = \sum_{m=1}^{M} \psi_m(x) w_m\,,$$

$$J(x) = \sum_{m=1}^{M} \mu_m \psi_m(x) w_m\,.$$

Method of characteristics

Let us introduce a spatial mesh

$$\{x_{i+1/2}, i = 0, ..., N, 0 = x_{1/2} < ... < x_{i+1/2} < ... < x_{N+1/2} = X\}\,.$$

We assume that cross sections and the source are piece-wise constant functions on a set of mesh cells $[x_{i-1/2}, x_{i+1/2}]$, i.e.

$$\Sigma_t(x) = \Sigma_{t,i}\,,\ \Sigma_s(x) = \Sigma_{s,i}\,,\ \Sigma_f(x) = \Sigma_{f,i}\,,$$
$$\nu_f(x) = \nu_{f,i}\,,\ q(x) = q_i\,,\quad \text{for}\ \ x_{i-1/2} < x < x_{i+1/2}\,. \tag{5}$$

Equations along characteristics

$$\psi(x_{i+1/2}, \mu_m) = \psi(x_{i-1/2}, \mu_m) e^{-\frac{1}{\mu_m}\int_{x_{i-1/2}}^{x_{i+1/2}} \Sigma_t(x)dx}$$
$$+ \frac{1}{\mu_m}\int_{x_{i-1/2}}^{x_{i+1/2}} Q(x') e^{-\frac{1}{\mu_m}\int_{x'}^{x_{i+1/2}} \Sigma_t(x'')dx''} dx'\,,\ \text{for}\ \mu_m > 0\,, \tag{6}$$

where

$$Q(x) = \frac{1}{2}\Big(\Sigma_s(x) + \nu_f(x)\Sigma_f(x)\Big)\phi(x) + \frac{1}{2}q(x)\,.$$

Method of characteristics

- A general scheme for $\psi_{m,i+1/2} = \psi(x_{i+1/2}, \mu_m)$ $(\mu_m > 0)$:

$$\psi(x_{i+1/2}, \mu_m) = \psi(x_{i-1/2}, \mu_m) e^{-\frac{\Sigma_{t,i}\Delta x_i}{\mu_m}} + \frac{1}{\mu_m} \int_{x_{i-1/2}}^{x_{i+1/2}} Q(x') e^{-\frac{\Sigma_{t,i}(x_{i+1/2} - x')}{\mu_m}} dx' \,, \quad (7)$$

$$\Delta x_i = x_{i+1/2} - x_{i-1/2} \,.$$

- Characteristic schemes differ from each other by approximation of the scalar flux and the total source term Q.
- If one approximates the total source as follows

$$Q(x) = Q_i = \frac{1}{2}\Big(\big(\Sigma_{s,i} + \nu_{f,i}\Sigma_{f,i}\big)\phi_i + q_i\Big) \,, \quad \text{for} \quad x_{i-1/2} < x < x_{i+1/2} \,,$$

where

$$\phi_i = \frac{1}{\Delta x_i} \int_{x_{i-1/2}}^{x_{i+1/2}} \phi(x) dx \quad \text{is the cell-average scalar flux} \,.$$

The characteristic equation (7) results in

$$\psi_{m,i+1/2} = \psi_{m,i-1/2} e^{-\tau_{m,i}} + \frac{Q_i}{\Sigma_{t,i}}\Big(1 - e^{-\tau_{m,i}}\Big) \,, \quad \text{for } \mu_m > 0 \,, \quad (8)$$

$$\tau_{m,i} = \frac{\Sigma_{t,i}\Delta x_i}{|\mu_m|} \,.$$

Concept of particle balance in discrete space

- It is important for a transport scheme to preserve *particle balance* in spatial cells.
- To derive the detailed balance equation in the ith cell, we integrate the transport equation (4) over $[x_{i-1/2}, x_{i+1/2}]$ to obtain

$$\mu_m(\psi_{m,i+1/2} - \psi_{m,i-1/2}) + \Sigma_{t,i}\Delta x_i \psi_{m,i} = \frac{\Delta x_i}{2}\Big((\Sigma_{s,i} + \nu_{f,i}\Sigma_{f,i})\phi_i + q_i\Big) \qquad (9)$$

$$\psi_{m,i} = \frac{1}{\Delta x_i}\int_{x_{i-1/2}}^{x_{i+1/2}} \psi_m(x)dx \quad \text{is the cell-average angular flux .}$$

- How to define the cell-average scalar flux ϕ_i =?
- We integrate Eq.(9) over $-1 \le \mu \le 1$ applying the quadrature sum

$$\sum_m w_m\Big[\mu_m(\psi_{m,i+1/2} - \psi_{m,i-1/2}) + \Sigma_{t,i}\Delta x_i \psi_{m,i}\Big] = \Delta x_i\Big((\Sigma_{s,i} + \nu_{f,i}\Sigma_{f,i})\phi_i + q_i\Big) .$$

- The cell-edge current $J_{i+1/2} = \sum_{m=1}^{M} \mu_m \psi_{m,i+1/2} w_m$

$$J_{i+1/2} - J_{i-1/2} + \Sigma_{t,i}\Delta x_i \sum_m w_m\psi_{m,i} = \Delta x_i\Big((\Sigma_{s,i} + \nu_{f,i}\Sigma_{f,i})\phi_i + q_i\Big) .$$

- The particle balance equation

$$J_{i+1/2} - J_{i-1/2} + (\Sigma_{t,i} - \Sigma_{s,i} - \nu_{f,i}\Sigma_{f,i})\Delta x_i\phi_i = q_i\Delta x_i .$$

- Thus, we need to define

$$\boxed{\phi_i = \sum_{m=1}^{M} \psi_{m,i} w_m .}$$

Step characteristics

The *step characteristics* (SC) method is defined by

- the discrete characteristic equations
- the auxiliary relation for the cell-average and cell-edge angular fluxes in the ith spatial cell

$$\psi_{m,i} = \alpha_{m,i}\psi_{m,i-1/2} + (1-\alpha_{m,i})\psi_{m,i+1/2}\,.$$

The weights $\alpha_{m,i}$ are defined in such a way that if

- $\Sigma_s = 0$ and $\Sigma_f = 0$,
- $q = 0$,

then the detailed balance equation result in the following exact relationships:

$$\psi_{m,i\pm1/2} = \psi_{m,i\mp1/2}e^{-\tau_{m,i}}\,, \quad \text{for } \mu_m \gtrless 0\,.$$

This leads to

$$\alpha_{m,i} = \pm\frac{1}{\tau_{m,i}} - \frac{e^{\mp\tau_{m,i}}}{1-e^{\mp\tau_{m,i}}} \quad \text{for} \quad \mu_m \gtrless 0\,.$$

Step characteristics

As a result, the equations of the SC method can be cast as follows:

$$\psi_{m,i-1/2} = \psi_{m,i+1/2} e^{-\tau_{m,i}} + \frac{Q_i}{\Sigma_{t,i}}(1 - e^{-\tau_{m,i}}), \quad i = N, ..., 1, \text{ for } \mu_m < 0,$$

$$\psi_{m,i+1/2} = \psi_{m,i-1/2} e^{-\tau_{m,i}} + \frac{Q_i}{\Sigma_{t,i}}(1 - e^{-\tau_{m,i}}), \quad i = 1, ..., N, \text{ for } \mu_m > 0,$$

$$\psi_{m,i} = \alpha_{m,i}\psi_{m,i-1/2} + (1 - \alpha_{m,i})\psi_{m,i+1/2}, \quad i = 1, ..., N,$$

$$\tau_{m,i} = \frac{\Sigma_{t,i}\Delta x_i}{|\mu_m|}$$

$$\alpha_{m,i} = \begin{cases} \frac{1}{\tau_{m,i}} - \frac{e^{-\tau_{m,i}}}{1-e^{-\tau_{m,i}}} & \text{for} \quad \mu_m > 0, \\ -\frac{1}{\tau_{m,i}} + \frac{1}{1-e^{-\tau_{m,i}}} & \text{for} \quad \mu_m < 0, \end{cases}$$

$$\phi_i = \sum_{m=1}^{M} \psi_{m,i} w_m,$$

$$\phi_{i-1/2} = \sum_{m=1}^{M} \psi_{m,i-1/2} w_m,$$

$$J_{i-1/2} = \sum_{m=1}^{M} \mu_m \psi_{m,i-1/2} w_m.$$

Step characteristics

It can be shown that the SC equations are equivalent to just the cell balance equation

$$\mu_m(\psi_{m,i+1/2} - \psi_{m,i-1/2}) + \Sigma_{t,i}\Delta x_i \psi_{m,i} = \frac{\Delta x_i}{2}\left((\Sigma_{s,i} + \nu_{f,i}\Sigma_{f,i})\phi_i + q_i\right)$$

with the auxiliary relation

$$\psi_{m,i} = \alpha_{m,i}\psi_{m,i-1/2} + (1 - \alpha_{m,i})\psi_{m,i+1/2}\,,$$

where $\alpha_{m,i}$ is defined by

$$\alpha_{m,i} = \begin{cases} \frac{1}{\tau_{m,i}} - \frac{e^{-\tau_{m,i}}}{1-e^{-\tau_{m,i}}} & \text{for} \quad \mu_m > 0\,, \\ -\frac{1}{\tau_{m,i}} + \frac{1}{1-e^{-\tau_{m,i}}} & \text{for} \quad \mu_m < 0\,, \end{cases}$$

and

$$\phi_i = \sum_{m=1}^{M} \psi_{m,i} w_m\,.$$

Linear characteristics (LC)

- The characteristic equation

$$\psi_{m,i+1/2} = \psi_{m,i-1/2} e^{-\frac{\Sigma_{t,i}\Delta x_i}{\mu_m}} + \frac{1}{\mu_m} \int_{x_{i-1/2}}^{x_{i+1/2}} \tilde{Q}_i(x') e^{-\frac{\Sigma_{t,i}(x_{i+1/2}-x')}{\mu_m}} dx' , \quad \mu_m > 0 \tag{10}$$

with a linear approximation of the total source term

$$\tilde{Q}_i(x) = Q_i + P_{1,i}(x)\hat{Q}_i , \quad \text{for} \quad x \in [x_{i-1/2}, x_{i+1/2}] .$$

$$P_{1,i}(x) = \frac{2(x - x_i)}{\Delta x_i} \quad \text{is Legendre polynomial of degree 1 ,} \tag{11}$$

$$x_i = \frac{1}{2}(x_{i+1/2} + x_{i-1/2}) , \tag{12}$$

$$Q_i = \frac{1}{2}\left((\Sigma_{s,i} + \nu_{f,i}\Sigma_{f,i})\phi_i + q_i\right) ,$$

$$\hat{Q}_i = \frac{1}{2}\left((\Sigma_{s,i} + \nu_{f,i}\Sigma_{f,i})\hat{\phi}_i + \hat{q}_i\right) ,$$

$$\hat{f}_i = \frac{3}{\Delta x_i} \int_{x_{i-1/2}}^{x_{i+1/2}} P_{1,i}(x) f(x) dx \quad \text{is the first spatial Legendre polynomial moment .}$$

Linear characteristics (LC)

- Unknowns: $\psi_{m,i+1/2}, \psi_{m,i}, \hat{\psi}_{m,i}$
- The characteristic equation

$$\psi_{m,i+1/2} = \psi_{m,i-1/2} e^{-\frac{\Sigma_{t,i}\Delta x_i}{\mu_m}} + \frac{1}{\mu_m} \int_{x_{i-1/2}}^{x_{i+1/2}} \tilde{Q}_i(x') e^{-\frac{\Sigma_{t,i}(x_{i+1/2}-x')}{\mu_m}} dx' , \quad \mu_m > 0$$

$$\tilde{Q}_i(x) = \frac{1}{2}(\Sigma_{s,i} + \nu_{f,i}\Sigma_{f,i})\left(\phi_i + P_{1,i}(x)\hat{\phi}_i\right) + \frac{1}{2}\left(q_i + P_{1,i}(x)\hat{q}_i\right) .$$

- The detailed balance equation

$$\mu_m(\psi_{m,i+1/2} - \psi_{m,i-1/2}) + \Sigma_{t,i}\Delta x_i \psi_{m,i} = \frac{\Delta x_i}{2}((\Sigma_{s,i} + \nu_{f,i}\Sigma_{f,i})\phi_i + q_i) .$$

- The first moment equation

$$3\mu_m(\psi_{m,i+1/2} + \psi_{m,i-1/2} - 2\psi_{m,i}) + \Sigma_{t,i}\Delta x_i \hat{\psi}_{m,i} = \frac{\Delta x_i}{2}\left((\Sigma_{s,i} + \nu_{f,i}\Sigma_{f,i})\hat{\phi}_i + \hat{q}_i\right) .$$

  This equation is obtained by integrating the transport equation (4) over the ith spatial cell with weight $P_{1,i}(x)$. Hence, we apply $\int_{x_{i-1/2}}^{x_{i+1/2}} P_{1,i}(x) \cdot dx$ to Eq. (4).

- The grid functions of the scalar flux

$$\phi_i = \sum_{m=1}^{M} \psi_{m,i} w_m , \quad \hat{\phi}_i = \sum_{m=1}^{M} \hat{\psi}_{m,i} w_m .$$

Diamond differencing (DD) scheme

- DD scheme is not a characteristic method.
- It is defined with the cell balance equation

$$\mu_m(\psi_{m,i+1/2} - \psi_{m,i-1/2}) + \Sigma_{t,i}\Delta x_i \psi_{m,i} = \frac{\Delta x_i}{2}\Big((\Sigma_{s,i} + \nu_{f,i}\Sigma_{f,i})\phi_i + q_i\Big)\,.$$

  where the cell-average angular flux is approximated as follows:

$$\psi_{m,i} = \frac{1}{2}(\psi_{m,i-1/2} + \psi_{m,i+1/2})\,.$$

- Thus, the DD auxiliary relation

$$\psi_{m,i} = \alpha_{m,i}\psi_{m,i-1/2} + (1-\alpha_{m,i})\psi_{m,i+1/2}\,, \quad \text{where} \quad \alpha_{m,i} = \frac{1}{2}\,.$$

- The cell-average scalar flux

$$\phi_i = \sum_{m=1}^{M} \psi_{m,i} w_m = \frac{1}{2}(\phi_{i-1/2} + \phi_{i+1/2})\,,$$

$$\phi_{i+1/2} = \sum_{m=1}^{M} \psi_{m,i+1/2} w_m\,.$$

- DD scheme for $\mu_m > 0$

$$\psi_{m,i+1/2} = \frac{1 - 0.5\tau_{m,i}}{1 + 0.5\tau_{m,i}}\psi_{m,i-1/2} + \frac{0.5\Big((\Sigma_{s,i} + \nu_{f,i}\Sigma_{f,i})\phi_i + q_i\Big)\Delta x_i}{(1 + 0.5\tau_{m,i})}\,.$$

Linear discontinuous Galerkin (DG) method

- The *linear discontinuous* (LD) method is formulated for $\psi_{m,i+1/2}, \psi_{m,i}, \hat{\psi}_{m,i}$.
- The detailed balance equation: $\int_{x_{i-1/2}}^{x_{i+1/2}} \cdot dx$ to Eq. (4) to obtain

$$\mu_m(\psi_{m,i+1/2} - \psi_{m,i-1/2}) + \Sigma_{t,i}\Delta x_i \psi_{m,i} = \frac{\Delta x_i}{2}((\Sigma_{s,i} + \nu_{f,i}\Sigma_{f,i})\phi_i + q_i)\,. \tag{13}$$

- The first moment equation: $\int_{x_{i-1/2}}^{x_{i+1/2}} P_{1,i}(x) \cdot dx$ (see Eq. (11)) to Eq. (4) to obtain

$$3\mu_m(\psi_{m,i+1/2}+\psi_{m,i-1/2}-2\psi_{m,i})+\Sigma_{t,i}\Delta x_i \hat{\psi}_{m,i} = \frac{\Delta x_i}{2}\left((\Sigma_{s,i}+\nu_{f,i}\Sigma_{f,i})\hat{\phi}_i+\hat{q}_i\right). \tag{14}$$

- To close the system of discrete equations (13) and (14), we approximate the angular flux in the ith cell as the following linear function:

$$\psi_m(x) = \psi_{m,i} + \frac{2}{\Delta x_i}(x - x_i)\hat{\psi}_{m,i} \quad \text{for } x_{i-1/2} \le x \le x_{i+1/2}\,. \tag{15}$$

  This is the finite-element expansion of the solution in Legendre basis functions.
- We use (15) in the upwinding procedure to evaluate $\psi_{m,i+1/2} = \psi_m(x_{i+1/2})$ for $\mu_m > 0$ and $\psi_{m,i-1/2} = \psi_m(x_{i-1/2})$ for $\mu_m < 0$ and derive the auxiliary relations in the ith cell:

$$\psi_{m,i\mp 1/2} = \begin{cases} \psi_{m,i} - \hat{\psi}_i\,, & \text{for } \mu_m < 0\,, \\ \psi_{m,i} + \hat{\psi}_i\,, & \text{for } \mu_m > 0\,. \end{cases} \tag{16}$$

Linear discontinuous Galerkin (DG) method

As a result, the LD method is defined by

$$\mu_m(\psi_{m,i+1/2} - \psi_{m,i-1/2}) + \Sigma_{t,i}\Delta x_i \psi_{m,i} = \frac{\Delta x_i}{2}((\Sigma_{s,i} + \nu_{f,i}\Sigma_{f,i})\phi_i + q_i)\,,$$

$$3\mu_m(\psi_{m,i+1/2} + \psi_{m,i-1/2} - 2\psi_{m,i}) + \Sigma_{t,i}\Delta x_i \hat{\psi}_{m,i} = \frac{\Delta x_i}{2}((\Sigma_{s,i} + \nu_{f,i}\Sigma_{f,i})\hat{\phi}_i + \hat{q}_i)\,,$$

$$\psi_{m,i\mp 1/2} = \begin{cases} \psi_{m,i} - \hat{\psi}_i\,, & \text{for } \mu_m < 0\,, \\ \psi_{m,i} + \hat{\psi}_i\,, & \text{for } \mu_m > 0\,, \end{cases}$$

$$i = 1, ..., N\,,$$

where

$$\phi_i = \sum_{m=1}^{M} \psi_{m,i} w_m\,,$$

is cell-average scalar flux,

$$\hat{\phi}_i = \sum_{m=1}^{M} \hat{\psi}_{m,i} w_m\,,$$

is the linear spatial moment of the scalar flux.

Linear DG method: alternative form

- The angular flux finite element expansion is defined by
$$\psi_{m,i}(x) = \psi_{L,m,i} b_{L,i}(x) + \psi_{R,m,i} b_{R,i}(x)\,.$$
- The Lagrange basis functions are given by
$$b_{L,i} = \frac{1}{\Delta x_i}(x_{i+1/2} - x)\,, \quad b_{R,i} = \frac{1}{\Delta x_i}(x - x_{i-1/2})\,.$$
- The transport equation (4) is integrated over i^{th} spatial cell with $b_{c,i}$, $c = L, R$. Hence, we apply $\int_{x_{i-1/2}}^{x_{i+1/2}} b_{L,i}(x) \cdot dx$ and $\int_{x_{i-1/2}}^{x_{i+1/2}} b_{R,i}(x) \cdot dx$ to Eq. (4) to get
$$\mu_m \Big(L_i^b \psi_{m,i}^b + L_i \psi_{m,i} \Big) + \Sigma_{t,i} \Delta x_i M_i \psi_{m,i} = \frac{1}{2} \Big(\Sigma_{s,i} + \nu_{f,i} \Sigma_{f,i} \Big) \Delta x_i M_i \phi_i + \frac{\Delta x_i}{2} M_i \mathbf{q}_i\,,$$
$$L_i^b = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \quad L_i = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}, \quad M_i = \frac{1}{6} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix},$$
$$\boldsymbol{\psi}_{m,i} = (\psi_{L,m,i}, \psi_{R,m,i})^T\,, \quad \boldsymbol{\psi}_{m,i}^b = (\psi_{L,m,i}^b, \psi_{R,m,i}^b)^T\,,$$
$$\boldsymbol{\phi}_i = (\phi_{L,i}, \phi_{R,i})^T\,, \quad \phi_{c,i}^n = \sum_m \psi_{c,m,i}^n w_m\,.$$
$$\boldsymbol{q}_i = (q_{L,i}, q_{R,i})^T\,.$$
- Upwinding conditions
$$\psi_{L,m,i}^b = \begin{cases} \psi_{L,m,i} & \text{if } \mu < 0\,, \\ \psi_{R,m,i-1} & \text{if } \mu > 0\,, \end{cases} \qquad \psi_{R,m,i}^b = \begin{cases} \psi_{L,m,i+1} & \text{if } \mu < 0\,, \\ \psi_{R,m,i} & \text{if } \mu > 0\,. \end{cases}$$

Upstream corner-balance (UCB) method

Corner balance methods are defined by means of

- the particle balance in each halves (corners) of the spatial cell
- approximations for relationship between exiting angular fluxes and corner-averaged values

We integrate the transport equation over the left half of the cell: $\int_{x_{i-1/2}}^{x_i} \cdot dx$

$$\mu_m \int_{x_{i-1/2}}^{x_i} \frac{\partial \psi}{\partial x} dx = \mu_m (\psi_{m,i} - \psi_{m,i-1/2}) \tag{17}$$

$$\int_{x_{i-1/2}}^{x_i} \Sigma_t \psi dx = \Sigma_{t,i} \psi_{L,m,i} \frac{\Delta x_i}{2} \qquad \int_{x_{i-1/2}}^{x_i} \Sigma_s \phi dx = \Sigma_{s,i} \phi_{L,i} \frac{\Delta x_i}{2} \tag{18}$$

Integration of the transport equation over the right half of the cell: $\int_{x_i}^{x_{i+1/2}} \cdot dx$ yields

$$\mu_m \int_{x_i}^{x_{i+1/2}} \frac{\partial \psi}{\partial x} dx = \mu_m (\psi_{m,i+1/2} - \psi_{m,i}) \tag{19}$$

$$\int_{x_i}^{x_{i+1/2}} \Sigma_t \psi dx = \Sigma_{t,i} \psi_{R,m,i} \frac{\Delta x_i}{2}, \qquad \int_{x_i}^{x_{i+1/2}} \Sigma_s \phi dx = \Sigma_{s,i} \phi_{R,i} \frac{\Delta x_i}{2} \tag{20}$$

Upstream corner-balance (UCB) method

The particle balance in the left and right halves (corners) of the ith cell

$$\mu_m(\psi_{m,i} - \psi_{m,i-1/2}) + \frac{1}{2}\Sigma_{t,i}\Delta x_i \psi_{L,m,i} = \frac{\Delta x_i}{4}\left((\Sigma_{s,i} + \nu_{f,i}\Sigma_{f,i})\phi_{L,i} + q_{L,i}\right),$$

$$\mu_m(\psi_{m,i+1/2} - \psi_{m,i}) + \frac{1}{2}\Sigma_{t,i}\Delta x_i \psi_{R,m,i} = \frac{\Delta x_i}{4}\left((\Sigma_{s,i} + \nu_{f,i}\Sigma_{f,i})\phi_{R,i} + q_{R,i}\right),$$

Cell-average value

$$\psi_{m,i} = \frac{1}{2}(\psi_{L,m,i} + \psi_{R,m,i})$$

Upwinding auxiliary conditions

$$\psi_{m,i-1/2} = \begin{cases} \psi_{L,m,i} & i = 1, \ldots, N \\ \psi_{in}^{-} & i = N+1 \end{cases} \qquad \mu < 0$$

$$\psi_{m,i+1/2} = \begin{cases} \psi_{R,m,i} & i = 1, \ldots, N \\ \psi_{in}^{+} & i = 0 \end{cases} \qquad \mu > 0$$

Source iterations for one-group transport problems

The method of source iterations (SI) for solving **fixed-source transport problems** is defined by the following algorithm:

$$\mu \frac{\partial \psi^{(\ell+1/2)}}{\partial x} + \Sigma_t \psi^{(\ell+1/2)} = \frac{1}{2}\left(\Sigma_s + \nu_f \Sigma_f\right)\phi^{(\ell)} + \frac{1}{2} q(x)\,,$$

$$\psi(0,\mu)^{(\ell+1/2)} = \psi_{in}^{+}(\mu) \quad \text{for } \mu > 0\,, \quad \psi(X,\mu)^{(\ell+1/2)} = \psi_{in}^{-}(\mu) \quad \text{for } \mu < 0\,.$$

Here ℓ is the iteration index.

$$\phi^{(\ell+1)} = \int_{-1}^{1} \psi^{(\ell+1/2)} d\mu\,, \quad J^{(\ell+1)} = \int_{-1}^{1} \mu \psi^{(\ell+1/2)} d\mu\,.$$

Basic iteration method for k-eigenvalue transport problem

In the case of k-**eigenvalue problems**, the SI method is defined by the following steps:

$$\mu \frac{\partial \psi^{(\ell+1/2)}}{\partial x} + \Sigma_t \psi^{(\ell+1/2)} = \frac{1}{2}\left(\Sigma_s + \frac{\nu_f \Sigma_f}{k^{(\ell)}}\right)\phi^{(\ell)}\,,$$

$$\psi(0,\mu)^{(\ell+1/2)} = 0 \quad \text{for } \mu > 0\,, \quad \psi(X,\mu)^{(\ell+1/2)} = 0 \quad \text{for } \mu < 0\,.$$

$$\phi^{(\ell+1)} = \int_{-1}^{1} \psi^{(\ell+1/2)} d\mu\,, \quad J^{(\ell+1)} = \int_{-1}^{1} \mu \psi^{(\ell+1/2)} d\mu\,.$$

$$k^{(\ell+1)} = \frac{\int_0^X \nu_f \Sigma_f \phi^{(\ell+1)} dx}{J^{(\ell+1)}(X) - J^{(\ell+1)}(0) + \int_0^X \Sigma_a \phi^{(\ell+1)} dx}$$

or

$$k^{(\ell+1)} = k^{(\ell)} \frac{\int_0^X \nu_f \Sigma_f \phi^{(\ell+1)} dx}{\int_0^X \nu_f \Sigma_f \phi^{(\ell)} dx}\,.$$